APPENDIX

A. Implementation Details

For all experiments, we set the following default parameters: memory sizes $|\mathbb{S}| = 64$ and $|\mathbb{L}| = 512$, a batch size of 64 for each replay including long-term and short-term relays, a regularization parameter $\lambda = 1.0$ for RR, a keyframe selection threshold $\delta_l = 0.02$, an id switch threshold $\delta_{\text{sw}} = 0.35$, a ReID threshold $\delta_{\text{reid}} = 0.7$ and a number of consecutive frames $\zeta_{\text{reid}} = 5$. In this paper, for representing the part-level features, we define ten parts: {front, back}×{head, torso, legs, feet, whole}.

For orientation estimation, we employ MonoLoc[1] to infer the orientation using detected joint positions from AlphaPose[2]. These joint positions are also utilized to estimate the visible parts. We utilize YOLOX-S[3] for bounding-box detection. For the tracking module, we utilize ByteTrack as our tracking method. For our proposed ReID module, we use ResNet18 as our feature extractor, pre-trained on the MOT16 dataset. During OCL for the ResNet18, only the layers after *conv3* are trainable (including *conv3*).

All evaluations are conducted on a computer with an Intel® Core™ i9-12900K CPU and NVIDIA GeForce RTX 3090. For real robot experiments, we use a Unitree Go1 quadruped robot with an Intel NUC 11 mini PC powered by Core i7-1165G7 CPU and NVIDIA GeForce RTX2060-laptop GPU. A dual-fisheye Ricoh camera is mounted on the robot, providing cropped perspective images with a resolution of 640×480 and a frequency of 30Hz.

B. OCLReID Lifecycle

Algorithm 1: ReID Lifecycle of OCLReID

Input: Current image $\mathbf{I}$ and tracked people $\{\mathbf{B}, \mathbf{p}\}_i$ representing bounding boxes and positions, target person's identity id, target confidence s, short-term memory $\mathbb{S}$, long-term memory $\mathbb{L}$, feature extractor f and target classifier g

Output: Target person's position $\{\mathbf{p}\}_{id}$ in the current frame

Extract image patches $\mathbf{M}$ from $\mathbf{I}$ and $\mathbf{B}$;
Construct the observation set $\{\mathbf{M}, \mathbf{y}\}_i$ where $\mathbf{y} = 1$ if $i == id$, otherwise $\mathbf{y} = 0$;
Extract features $\mathbf{F}$ from $\mathbf{M}$ with f;
if $id \in \{i\}$ **then**
  Estimate s of the target person based on Eq. 4;
  if $s > \delta_{\text{sw}}$ **then**
    Consider $\{\bar{i}\}$ as identities of negative tracks;
    $\{\mathbf{M}, \mathbf{y}\}_{id} \rightarrow \mathbb{L}$ if it is a keyframe based on Eq. 5;
    $\{\mathbf{M}, \mathbf{y}\}_{\bar{i}} \rightarrow \mathbb{L}$ based on FILO rule;
    Consolidate $\mathbb{L}$ with OCL techniques if $\mathbb{L}$ is full;
    $\{\mathbf{M}, \mathbf{y}\}_{id} \rightarrow \mathbb{S}$, $\{\mathbf{M}, \mathbf{y}\}_{\bar{i}} \rightarrow \mathbb{S}$ based on FILO rule;
    Sample m_{st} and m_{lt} from $\mathbb{S}$ and $\mathbb{L}$, respectively;
    Train f with m_{st} and m_{lt} based on Eq. 1;
    Train g with m_{st} based on Eq. 3;
    Return target position $\{\mathbf{p}\}_{id}$;
  else
    Let $id = -1$, indicates id switch between the target person and other people;
    Return ø;
else
  Estimate s of the i_{th} person based on Eq. 4;
  if $s > \delta_{\text{reid}}$ for consecutive ζ_{reid} frames **then**
    Let $id = i$, indicates successful target person ReID;
    Return target person's position $\{\mathbf{p}\}_{id}$;
  else
    Return ø;

[1] https://github.com/vita-epfl/monoloco
[2] https://github.com/MVIG-SJTU/AlphaPose
[3] https://github.com/Megvii-BaseDetection/YOLOX

C. Memory Examples

Several replayed examples of our short-term and long-term memories are shown in Fig. 5. For more visual examples of short-term and long-term memories, please refer to the supplementary video.

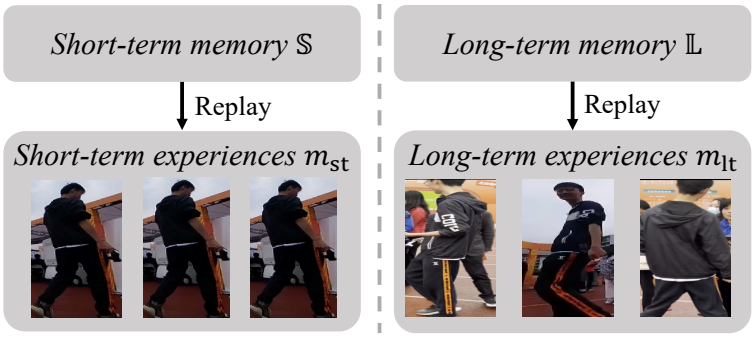

Fig. 5. Both short-term and long-term experiences $(m_{\mathrm{st}} \cup m_{\mathrm{lt}})$ are responsible for training the feature extractor with m_{st} and m_{lt} being sampled from short-term memory $\mathbb{S}$ and long-term memory $\mathbb{L}$, respectively. $\mathbb{L}$ contains sparse yet valuable historical samples, maintained by the *memory manager*. Besides, the target classifier is trained with m_{st} sampled from $\mathbb{S}$, which stores the most recently observed samples, representing the latest knowledge.

D. Example Images from Self-built Dataset

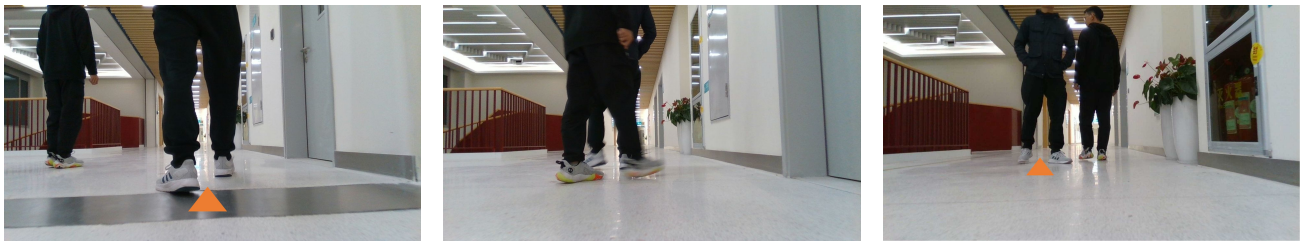

Fig. 6. Examples of the custom-built dataset. It contains challenging situations of similar appearance of distracting people and different appearance after occlusion (lower back body vs. whole front body).